

## **CAM6134-T2610 Assignment**

### **Simulation of a Hospital Emergency Department Queuing System**

**Project Members:** 3-4 members (within the same tutorial class)

**Deadline To Submit:** 22 June 2026, 23:59 (Week 13)

**Presentation, Q&A:** 15 minutes per group, physical presentation.

#### **1. Background**

In a hospital emergency department, patients arrive randomly and wait to be treated by doctors. During busy periods, long waiting times may occur. The hospital aims to evaluate and improve its service system using simulation modelling.

#### **2. Objective**

This assignment aims to enable students to:

- Apply fundamental queuing theory concepts in the development of a simulation model
- Develop and implement a functional simulation using Octave or FreeMat
- Analyse system performance using relevant performance metrics
- Propose and justify system improvements based on simulation results

#### **3. General Requirements**

Students must:

- Build a working simulation model using Octave or FreeMat
- Define and justify arrival and service processes
- Implement an appropriate queue structure (single-server, multi-server, or priority)
- Compute key performance metrics
- Compare at least one improvement scenario

#### **4. Performance Metrics**

Students must compute and analyse:

- Average waiting time
- Average queue length
- Doctor utilization
- Total patients served

## 5. Deliverables

- Report (5–10 pages in **pdf format**)
- Simulation code (Octave / FreeMat) (**.m file** and **readme file**)
- Presentation (5 minutes) & Q&A (5 minutes) (max 10 slides in **pdf format**)
- Marks will be deducted for late submissions.

## 6. Assessment Rubric (Scale 1–4)

### 6.1 Simulation Model (12 marks)

#### Component A: Model Design & Implementation (4 marks)

| Level            | Description                                                                                                                      |
|------------------|----------------------------------------------------------------------------------------------------------------------------------|
| 4 (Excellent)    | Implements a priority queue model with multiple doctors. Correct priority handling and queue logic. Fully functional and stable. |
| 3 (Good)         | Implements a multi-server model (M/M/c) with 2–3 doctors and FIFO queue. Model is functional and supports scenario comparison.   |
| 2 (Satisfactory) | Implements a basic single-server model (M/M/1) with FIFO queue, random arrivals, and service times.                              |
| 1 (Poor)         | Simulation is incomplete, incorrect, or not functioning.                                                                         |

#### Component B: Arrival and Service Modelling (4 marks)

| Level            | Description                                                                                                                                                                                                                                                                                                   |
|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 4 (Excellent)    | Clearly defines and applies appropriate arrival and service processes (e.g., Poisson arrivals, exponential service times or justified alternatives). Parameters are realistic and well-justified (e.g., peak vs non-peak rates). Demonstrates clear understanding of how assumptions affect system behaviour. |
| 3 (Good)         | Uses appropriate arrival and service distributions with reasonable parameter values. Some justification provided, but explanation lacks depth.                                                                                                                                                                |
| 2 (Satisfactory) | Uses basic or arbitrary random data generation with minimal or unclear assumptions. Limited justification.                                                                                                                                                                                                    |
| 1 (Poor)         | Arrival and service processes are incorrect, missing, or not aligned with queuing concepts. No clear assumptions.                                                                                                                                                                                             |

**Component C: Results Interpretation & Functionality (4 marks)**

| Level            | Description                                                                                                                                 |
|------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| 4 (Excellent)    | Accurately computes performance metrics and clearly interprets results. Demonstrates ability to compare scenarios and justify improvements. |
| 3 (Good)         | Computes metrics correctly with some interpretation. Limited comparison or justification.                                                   |
| 2 (Satisfactory) | Basic results presented with minimal explanation.                                                                                           |
| 1 (Poor)         | Results missing, incorrect, or not interpreted.                                                                                             |

**6.2 Documentation (Report) (4 marks)**

The report should clearly document the application of simulation and analysis.

**Expected Components:**

- Problem description and objectives
- Assumptions (arrival/service processes)
- Queuing model description
- Simulation methodology
- Results (tables/graphs)
- Analysis and discussion
- Conclusion and recommendations

| Level            | Description                                                                                                                                          |
|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| 4 (Excellent)    | Complete, well-structured, and clearly explains methodology, application, and results. Strong analysis with meaningful insights and recommendations. |
| 3 (Good)         | Generally clear with most components included. Some sections lack depth.                                                                             |
| 2 (Satisfactory) | Basic components present but lacks clarity or completeness.                                                                                          |
| 1 (Poor)         | Incomplete or poorly organized. Missing major components.                                                                                            |

### 6.3 Assignment Presentation (4 marks)

| Level            | Description                                                                                                                                            |
|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| 4 (Excellent)    | Clear, logical explanation of model, assumptions, and results. Demonstrates strong understanding and answers questions confidently with justification. |
| 3 (Good)         | Generally clear explanation with good understanding. Minor gaps in reasoning.                                                                          |
| 2 (Satisfactory) | Basic explanation with limited clarity and weak justification.                                                                                         |
| 1 (Poor)         | Unclear or disorganized presentation. Weak understanding and inability to justify work.                                                                |

### 7. Academic Integrity (Plagiarism Policy)

All submitted work must be original. Students must adhere to the university's academic integrity guidelines.

- Similarity checks may be conducted
- Projects with >50% similarity may be penalized
- Penalties include mark deduction, zero marks, or disciplinary action

Students must ensure:

- Work is independently developed within the group
- All sources are properly cited